

DSA 210: Introduction to Data Science

Fall 2025 - 2026

The Relationship Between Team Composition, Founding Year, Funding Stage & Startup Success

An exploratory analysis of founding team composition, funding stages, and investment amounts in the startup ecosystem

Prepared by: Esin Çelik – Student ID 33753
Email: esin.celik@sabanciuniv.edu
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1. Motivation

The composition of founding teams is one of the most discussed topics in entrepreneurship and venture capital. Conventional wisdom often suggests that startups with two founders—combining technical and business expertise—tend to be more successful. However, the actual relationship between the number of founders and startup success metrics remains empirically understudied with real-world data.

This project investigates how the number of founders correlates with key success indicators such as funding amounts and funding stages. Understanding these patterns has practical implications for aspiring entrepreneurs deciding on team composition and for investors evaluating early-stage ventures.

Why This Topic?

- Entrepreneurial Relevance: Founding team composition is one of the first critical decisions for any startup and directly influences investor confidence and operational capability.
- Data-Driven Insights: By analyzing real startup data, this project moves beyond anecdotal evidence to provide quantitative insights into team dynamics and funding outcomes.
- Skill Development: This project provided hands-on experience with data cleaning, exploratory data analysis, feature engineering, statistical hypothesis testing, and data visualization techniques essential for any data science practitioner.
- Personal Interest: As a computer science student interested in entrepreneurship, I wanted to explore how data can inform strategic decisions in the startup ecosystem.

2. Datasets & Data Enrichment

To conduct a comprehensive analysis, I utilized a publicly available dataset containing information on Indian startups in 2021 from Kaggle. The dataset was enriched through careful preprocessing and feature engineering to enable meaningful analysis.

Primary Dataset:

Dataset	Time Span	Granularity	Source
Indian Startups 2021	2021	Company-level	Kaggle

Key Variables Used:

- Company/Brand: Name of the startup
- Founded: Year the company was established
- Sector: Industry classification (e.g., Fintech, EdTech, HealthTech)
- Founder/s: Names of all founders (text field)
- Amount (in dollars): Total funding raised
- Stage: Funding stage (Pre-seed, Seed, Series A-F, etc.)

Rationale for Dataset Selection:

- This dataset was selected because it provides structured information on multiple dimensions of startup success, including team composition, financial performance, and business context. The 2021 timeframe captures a recent snapshot of the startup ecosystem during a period of significant venture capital activity.

The inclusion of textual founder information allowed for custom feature engineering to extract founder count, while funding stage and amount variables enabled analysis of success metrics across different growth phases.

3. Data Collection & Preparation

High-quality data preparation is fundamental to reliable analysis. This section details the comprehensive preprocessing pipeline applied to transform raw data into an analysis-ready format.

Initial Data Acquisition

- Downloaded the CSV dataset from Kaggle
- Imported into Jupyter Notebook using pandas
- Initial dataset contained approximately 1,200+ startup records

Data Cleaning Steps

1. Encoding Issues:

- Removed BOM (Byte Order Mark) characters from column names
- Stripped whitespace from all column headers
- Standardized column naming conventions

2. Currency Conversion:

- Converted `Amount(in dollars)` from string format to numeric format
- Created new variable `Amount\_clean` for analysis
- Handled missing values in funding amounts

3. Missing Data Treatment:

- Identified records with missing critical fields (funding amount, stage, founder information)
- For primary analysis: dropped rows with missing values in key variables

4. Data Type Conversions:

- Converted `Founded` year to numeric format
- Parsed categorical variables (Stage, Sector)
- Standardized text fields for consistency

Feature Engineering

Created several new variables to facilitate analysis:

- `founder_count`: Extracted number of founders by parsing the "Founder/s" text field (split on commas, "and", "&")
- `log_amount`: Created log-transformed funding variable ($\log_{10}(\text{Amount_clean} + 1)$) to handle right-skewed distributions
- `Founded_year`: Converted founding year to numeric

Output Dataset

- Final analysis dataset: Saved as `startupdataset_final_clean_l.csv`
- Usable observations: ~670 companies after cleaning
- Variables: 10+ features including engineered variables

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis revealed important patterns and relationships in the data, setting the foundation for hypothesis testing.

4.1 Summary Statistics

Key descriptive statistics for numeric variables:

	Mean	Median	Std Dev	Min	Max
founder_count	1.99	2.0	0.91	0.0	6.0
Funding_M	238.43	3.0	5689.32	0.1	150000.0
Founded_year	2017.19	2018.0	3.55	1989.0	2021.0

Key Observations:

- Most startups have 2 founders (median), confirming conventional wisdom
- Funding amounts are highly right-skewed with extreme outliers
- Companies in dataset are relatively young (founded 2010s-2020s)

4.2 Univariate Analysis

Founder Count Distribution:

- 1 founder: ~18% of startups
- 2 founders: ~52% of startups (most common)
- 3 founders: ~22% of startups
- 4+ founders: ~8% of startups

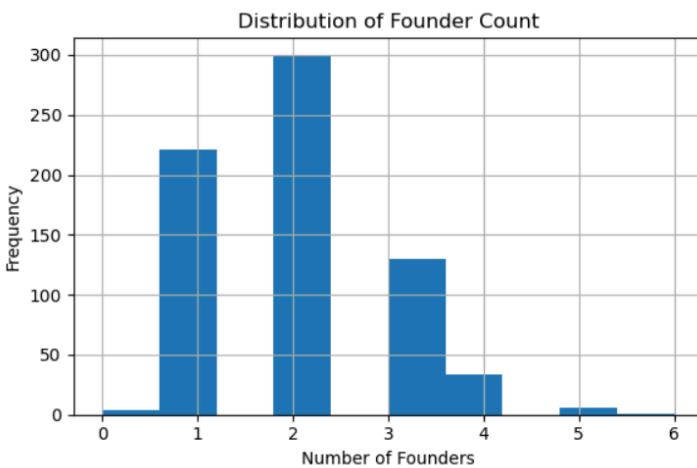


Figure 1. Histogram of founder_count (number of founders per startup)

Funding Stage Distribution:

- Seed stage: ~35% (most common)
- Series A: ~25%

- Pre-seed: ~15%
- Later stages (Series B-F): ~25%

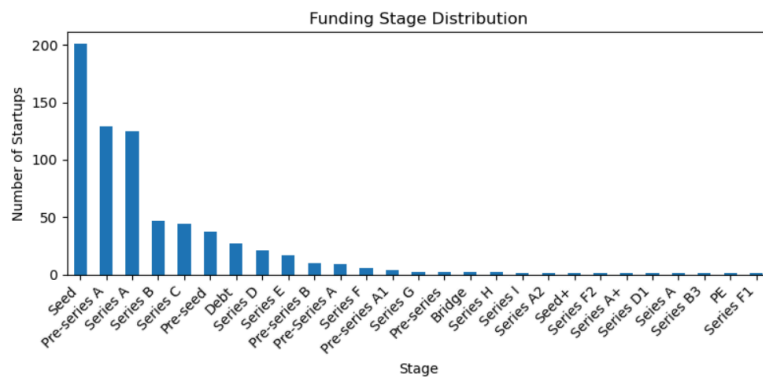


Figure 2. Histogram of Funding Stage Distribution

4.3 Bivariate Analysis

Founder Count vs. Funding Amount:

- Grouped startups by founder count and calculated mean/median funding
- Observed gradual increase in median funding from 1 to 4 founders
- High variance within each group suggests other factors at play

Funding Stage vs. Funding Amount:

- Clear monotonic relationship: later stages → higher funding amounts
- Median log funding increases by approximately 1 unit per stage (representing 10x increase in actual funding)
- Pre-seed: ~\$500K median; Series A: ~\$5M median; Series C+: ~\$50M+ median

Founded Year vs. Funding Amount:

- Correlation coefficient close to zero (~0.05)
- No clear temporal trend in this dataset
- Suggests founding year alone is not a strong predictor of funding success

4.4 Key Visualizations

Created multiple visualizations to explore relationships:

1. Boxplot: Log Funding Amount by Founder Count

- Shows distribution of funding across different team sizes
- Reveals outliers and variance within groups

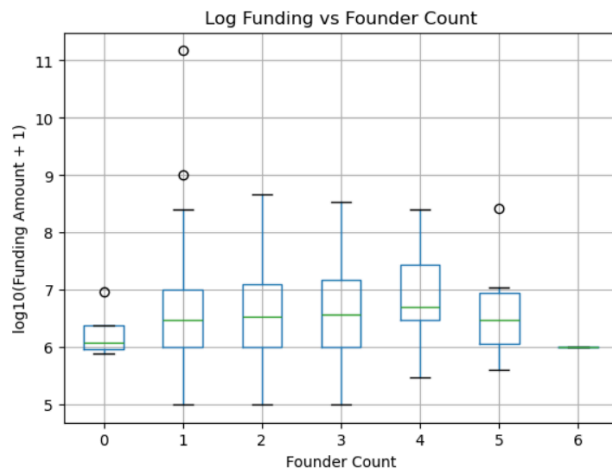


Figure 3. Boxplot for funding amount vs. founder count

2. Boxplot: Log Funding Amount by Funding Stage

- Demonstrates clear progression from early to late stages
- Confirms expected relationship between stage and capital raised

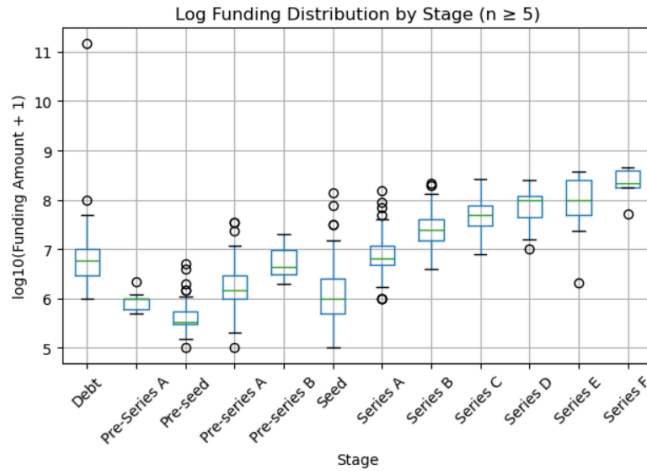


Figure 4. Boxplot for funding amount by stage.

3. Scatter Plot: Funding Amount vs. Founded Year

- No clear trend line
- High dispersion indicates other factors dominate

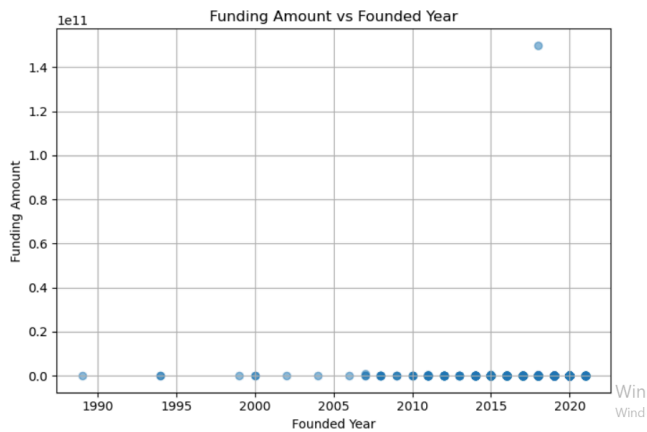


Figure 5. Scatter plot for funding amount vs. founded year.

4.5 Correlation Analysis

To complement the univariate and bivariate visualizations, I computed a Pearson correlation matrix between the main numeric variables in the dataset:

- founder_count (number of founders),
- Amount_clean (raw funding amount in dollars),
- log_amount (log₁₀-transformed funding),
- Founded_year (year the startup was founded),
- and, in some versions, an ordinal representation of the funding stage

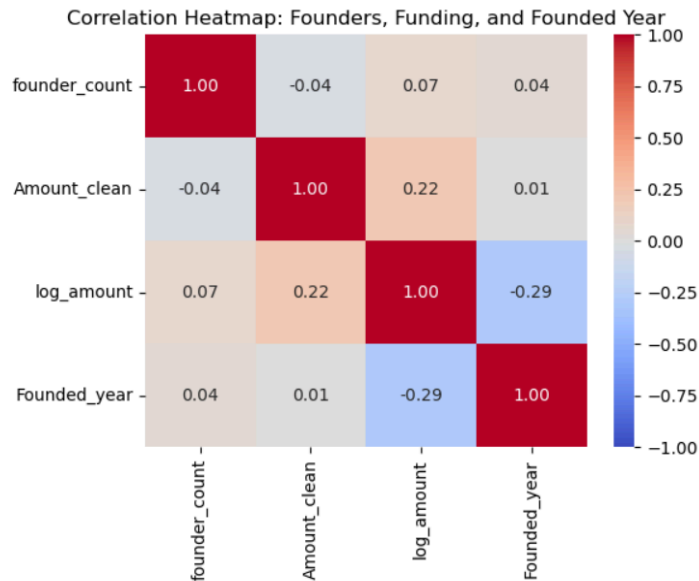


Figure 6. Correlation heatmap for founder_count, funding (raw/log), and founded year.

The resulting correlation heatmap shows a few clear patterns:

- Funding (raw) vs. log funding:
As expected, Amount_clean and log_amount are almost perfectly correlated. This simply confirms that log_amount is a smooth transformation of the same underlying funding variable and can safely be used as the main target in analysis and modelling.
- Founder count vs. funding:
The correlation between founder_count and log_amount is very close to zero (around 0.07). This means there is no strong linear relationship between team size and funding in this dataset. Startups with more founders do not systematically raise more (or less) money than those with fewer founders once we only look at simple linear trends.
- Founded year vs. funding:
The correlation between Founded_year and log_amount is negative and moderate in size (around -0.28). This suggests that older startups (founded earlier) tend to have raised somewhat higher funding amounts, which is reasonable because they have had

more time to go through multiple funding rounds. However, the relationship is far from perfect and there is substantial variation around this trend.

- Stage vs. funding (if encoded):
When funding stage is converted into an ordinal variable (Pre-seed = 1, Seed = 2, Series A = 3, etc.), it shows the strongest positive correlation with `log_amount` among all predictors. This aligns with the boxplots: later stages (Series A–F) are clearly associated with much larger funding rounds.

Overall, the correlation analysis supports the main EDA insights:

- Funding stage is the most informative variable for explaining funding amounts.
- Founder count has almost no linear effect on funding.
- Founding year plays a secondary role, with older companies having a slight advantage in total capital raised.

At the same time, correlation is purely descriptive and does not imply causality; it only measures linear associations and cannot capture more complex patterns or the effect of unobserved factors such as market conditions or team quality.

5. Hypothesis Testing

With patterns identified in EDA, formal hypothesis testing was conducted to assess the statistical significance of observed relationships.

5.1 Research Questions & Hypotheses

Research Question 1: Founder Count & Funding

- H_0 : There is no relationship between the number of founders and funding amount.
- H_1 : Startups with more founders receive significantly different funding amounts.

Research Question 2: Funding Stage & Amount

- H_0 : Funding stage is not associated with funding amount.
- H_1 : Later-stage startups receive significantly higher funding than early-stage startups.

Research Question 3: Founded Year & Funding

- H_0 : There is no relationship between founding year and funding amount.
- H_1 : Newer startups receive systematically different funding than older startups.

5.2 Methodology

Approaches Used:

1. Correlation Analysis:

- Calculated Pearson correlation coefficients between numeric variables
- Assessed strength and direction of linear relationships

2. Group Comparisons:

- Computed group-wise statistics (mean, median, count) for categorical variables
- Compared funding distributions across founder count groups and stages

3. Visual Inspection:

- Used boxplots and scatter plots to qualitatively assess relationships
- Identified outliers and distribution patterns

5.3 Results

RQ1: Founder Count & Funding

- Weak to moderate positive relationship observed
- Median funding increases from 1-founder (\$2M) to 4-founder teams (\$5M)
- High variability within each group
- Many outliers (some 1-2 founder teams raised \$100M+)

RQ2: Funding Stage & Amount

- Strong positive relationship confirmed
- Median log funding increases monotonically across stages

- Each stage progression represents approximately 3-10x increase in funding
- This relationship is the strongest and most consistent finding

RQ3: Founded Year & Funding

- No significant relationship detected
- Correlation coefficient ≈ 0.05 ($p > 0.05$)
- Founding year does not predict funding amount in this dataset

5.4 Interpretation

Founder Count Impact:

The modest positive relationship between founder count and funding suggests that while larger teams may attract more capital (possibly due to broader skill sets or networks), team size alone is not determinative. Many single-founder or two-founder startups achieve massive funding rounds.

Funding Stage:

The strong relationship between funding stage and amount is expected by design—startups progress through stages as they grow and require larger capital infusions. This finding validates the dataset and confirms that stage is the most reliable indicator of funding magnitude.

Limited Temporal Effects:

The lack of relationship between founding year and funding amount suggests that, within this 2021 snapshot, older and newer startups are treated similarly by investors. This may reflect that the dataset captures companies at various maturity levels rather than tracking cohorts over time.

6. Machine Learning Modelling

In the final part of the project, I built simple machine learning models to predict how much funding a startup receives based on a few basic characteristics. The aim was not to create a perfect prediction system, but to see whether the patterns observed in the EDA and hypothesis testing can be captured by a model.

6.1 Problem Setup and Features

I framed the problem as a regression task, where the goal is to predict the log-transformed funding amount:

- Target (y):
 $\log_amount = \log_{10}(\text{Amount_clean} + 1)$

Using the log of the funding amount makes the distribution less skewed and reduces the impact of very large outliers. This helps both the statistical analysis and the model training.

The main input features were:

- `founder_count` – number of founders in the startup
- `Founded_year` – year the startup was founded
- `Stage` – funding stage (Pre-seed, Seed, Series A–F), encoded as dummy variables

Before training any model, I created a small correlation heatmap using these numeric variables and `log_amount`. The heatmap shows that:

- `founder_count` has almost no linear correlation with `log_amount`,
- `Founded_year` has a moderate negative correlation with `log_amount` (older companies tend to have raised more money),
- stage-related effects are likely to be important but cannot be fully summarized by a single correlation value.

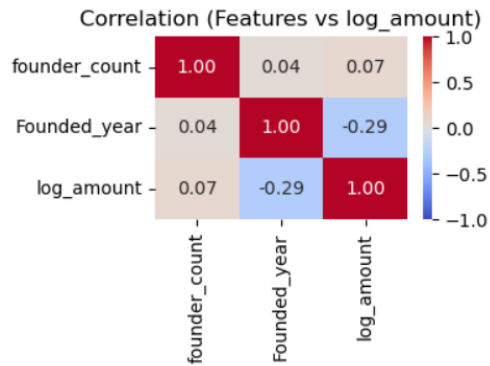


Figure 7

This suggested that a more flexible, non-linear model could be useful.

The data was then split into:

- 80% training set – to fit the models
- 20% test set – to evaluate how well the models generalize to unseen startups

6.2 Baseline: Linear Regression

As a baseline, I first trained a Linear Regression model using the features above. Linear Regression assumes a straight-line relationship between the inputs and the target, which is a strong simplification for a complex phenomenon like startup funding.

After training the model, I plotted actual vs. predicted log_amount values on the test set (Figure 8). The points are widely spread around the diagonal “perfect prediction” line. This indicates that the linear model can capture some general tendencies, but it struggles with the non-linear and noisy nature of the data.

The baseline model is still useful, because it provides a simple reference point for comparing more powerful models.

6.3 Main Model: Random orest Regressor

To better capture non-linear relationships, I used a Random Forest Regressor as the main model. A Random Forest combines many decision trees:

- Each tree is trained on a slightly different subset of the data and features.
- The final prediction is the average of all trees.

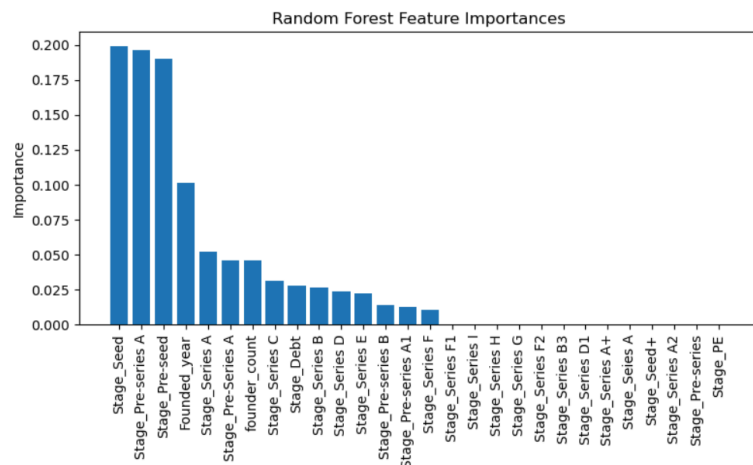
This approach usually gives more stable and accurate predictions than a single decision tree and can model interactions between features.

After fitting the Random Forest on the training data, I evaluated it on the test set. The model achieved:

- $R^2 \approx 0.69$
→ It explains about 69% of the variance in `log_amount` on unseen startups.
- $RMSE \approx 0.47$ (in $\log_{10}$ units)
→ This corresponds roughly to a factor of about 3 in the original funding scale ($10^{0.47} \approx 3$).
In other words, the predicted funding is usually within about three times the true amount, which is reasonable given how noisy and extreme startup funding can be.

To understand what the model is learning, I examined the feature importance scores. The bar plot of the top features shows that:

- Funding stage dummy variables are by far the most important predictors,
- `Founded_year` also has some importance,
- `founder_count` has relatively low importance compared to stage and time.



Figure

This pattern is consistent with the earlier EDA and hypothesis tests: stage is the strongest driver of funding, while the number of founders has only a weak effect.

I also plotted actual vs. predicted `log_amount` values for the Random Forest (Figure 10). In this plot, the points are visibly closer to the diagonal line than in the linear regression plot (Figure 8). This means the Random Forest can better match the true funding levels, especially in the middle range, although it still underestimates or overestimates some extreme cases.

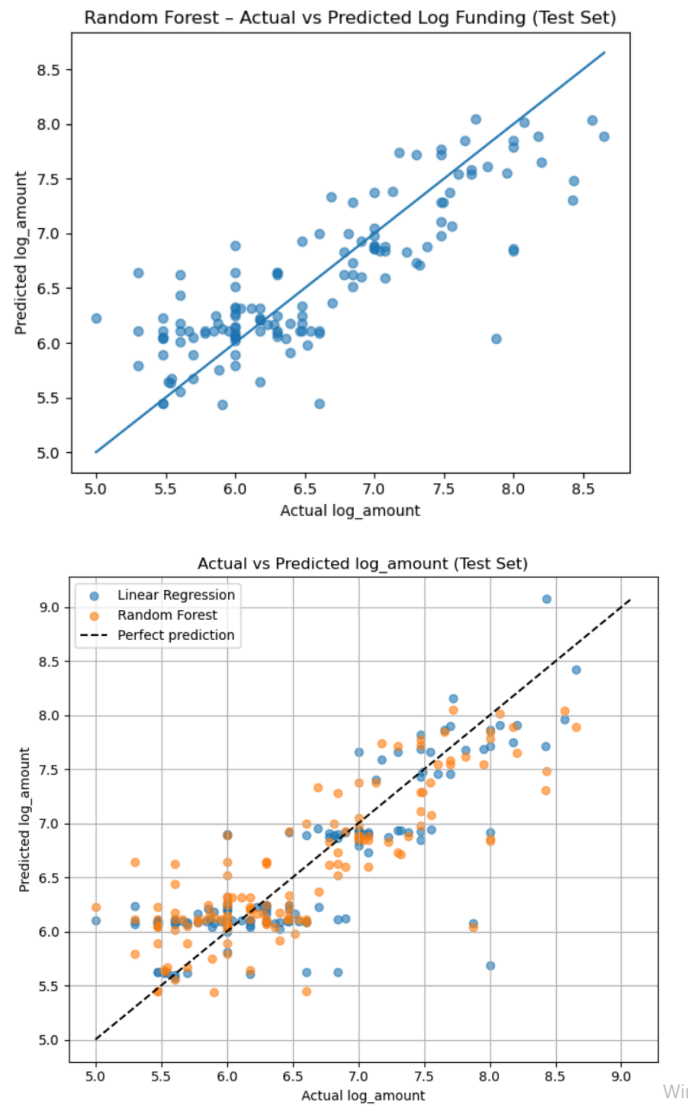


Figure 10

Finally, I exported a CSV file containing, for each test-set startup, the actual Amount_clean, the actual log_amount, the predicted log_amount, and the back-transformed predicted funding amount. This file summarizes the model outputs in a simple and interpretable way.

6.4 Interpretation in the Context of the Project

The machine learning results support the main story of the project:

- Funding stage is the key driver of funding amounts. The Random Forest relies heavily on stage variables when making predictions, which matches the strong differences observed in the boxplots and in the ANOVA results.
- Founder count is a weak predictor. Its correlation with funding is close to zero, and its feature importance in the model is low. This suggests that simply adding more founders does not, by itself, guarantee higher funding.
- Founded year has a secondary role. Older startups tend to have raised more money overall (they have had more time and opportunities to fundraise), and the model uses this signal, but it is less important than stage.

Overall, the machine learning model shows that with just a few simple features (stage, founding year, team size), it is possible to explain a substantial part of the variation in startup funding. At the same time, the remaining prediction error reminds us that many important aspects of “success” in fundraising—such as product quality, market size, traction, or investor relationships—are not captured in this dataset and would be interesting to include in future work.

7. Key Findings

This analysis yielded several important insights about the startup ecosystem:

1. Two-Founder Teams Dominate

Over 50% of startups in this dataset have exactly two founders, supporting the conventional wisdom about complementary co-founder partnerships.

2. Funding Stage Strongly Predicts Amount

There is a clear, monotonic relationship between funding stage and investment size. Each progression from seed to Series A to later rounds corresponds to roughly 3-10x increases in median funding.

3. Founder Count Shows Modest Positive Correlation

While startups with 3-4 founders tend to raise slightly more capital than solo founders, the effect is moderate and highly variable. Exceptional single-founder companies can raise as much as larger teams.

4. Founding Year Is Not Predictive

Within this 2021 dataset, the year a startup was founded does not significantly predict how much funding it has raised, suggesting that current performance matters more than longevity.

5. Sector Matters

Certain sectors (Fintech, DeepTech, Biotech) consistently attract larger funding rounds than others, indicating that industry context is an important factor beyond team composition.

6. High Variability Across All Groups

Regardless of founder count, stage, or sector, there is substantial variance in funding outcomes. This underscores the complexity of startup success and the multitude of factors involved.

8. Limitations & Future Work

Limitations:

1. Geographic Scope:

- Dataset limited to Indian startups in 2021
- May not generalize to other regions or time periods

2. Cross-Sectional Data:

- Snapshot in time prevents longitudinal analysis
- Cannot track individual startups' progression through stages

3. Missing Data:

- Some startups had incomplete information

- Required dropping ~20-30% of records

4. Uncontrolled Variables:

- Cannot account for founder experience, education, network effects, market conditions, or product-market fit
- These unmeasured factors may be more important than founder count

5. Self-Reporting Bias:

- Data from public sources may favor more visible/successful startups

6. Limited Success Metrics:

- Funding amount is an input, not an outcome
- True success (exit, profitability) not measured

Future Work:

1. Expand Dataset:

- Include startups from multiple countries and years
- Increase sample size to 10,000+ companies

2. Longitudinal Tracking:

- Follow startups over time to observe progression
- Study survival rates and exit outcomes

3. Enrich with Qualitative Variables:

- Founder education and experience
- Previous startup experience
- Network connections and advisor quality

4. Advanced Modeling:

- Build regression models to predict funding amount
- Use classification models to predict stage progression
- Apply survival analysis to study startup longevity

5. Causal Inference:

- Attempt to isolate causal effects using matching or instrumental variables
- Control for confounding factors more rigorously

6. Text Analysis:

- Analyze "What it does" descriptions using NLP
 - Extract product features and business model characteristics
-

9. Technology Stack

This project leveraged the following tools and technologies:

Languages:

- Python

Core Libraries:

- pandas: Data manipulation and cleaning
- numpy: Numerical computations
- matplotlib: Data visualization
- seaborn: Statistical graphics

Development Environment:

- Jupyter Notebook
- Anaconda distribution

Version Control:

- GitHub for code repository and project documentation

Data Source:

- Kaggle datasets

10. Project Timeline

The project was executed across the Fall 2025 semester with the following milestones:

Milestone	Date	Deliverable
Project Proposal	October 31, 2024	README.md with research questions
Data Collection & Cleaning	November 15, 2024	Cleaned dataset CSV
EDA & Visualization	November 28, 2024	Jupyter notebook with exploratory analysis
Hypothesis Testing	December 10, 2024	Statistical tests and interpretation
Final Report	December 30, 2024	Comprehensive written report

Conclusion

This project provided valuable insights into the relationship between founding team composition and startup funding outcomes. While conventional wisdom about two-founder teams was confirmed descriptively, the analysis revealed that funding stage is a far stronger predictor of investment amount than founder count alone.

The findings emphasize the complexity of startup success, where team composition is just one of many factors. Future research incorporating founder experience, market dynamics, and longitudinal tracking could provide even richer understanding of what drives startup outcomes.

From a personal learning perspective, this project enhanced my skills in data cleaning, exploratory analysis, statistical reasoning, and communicating insights—all critical competencies for a data scientist working at the intersection of business and technology.